PBMS2000
for Telecom Site



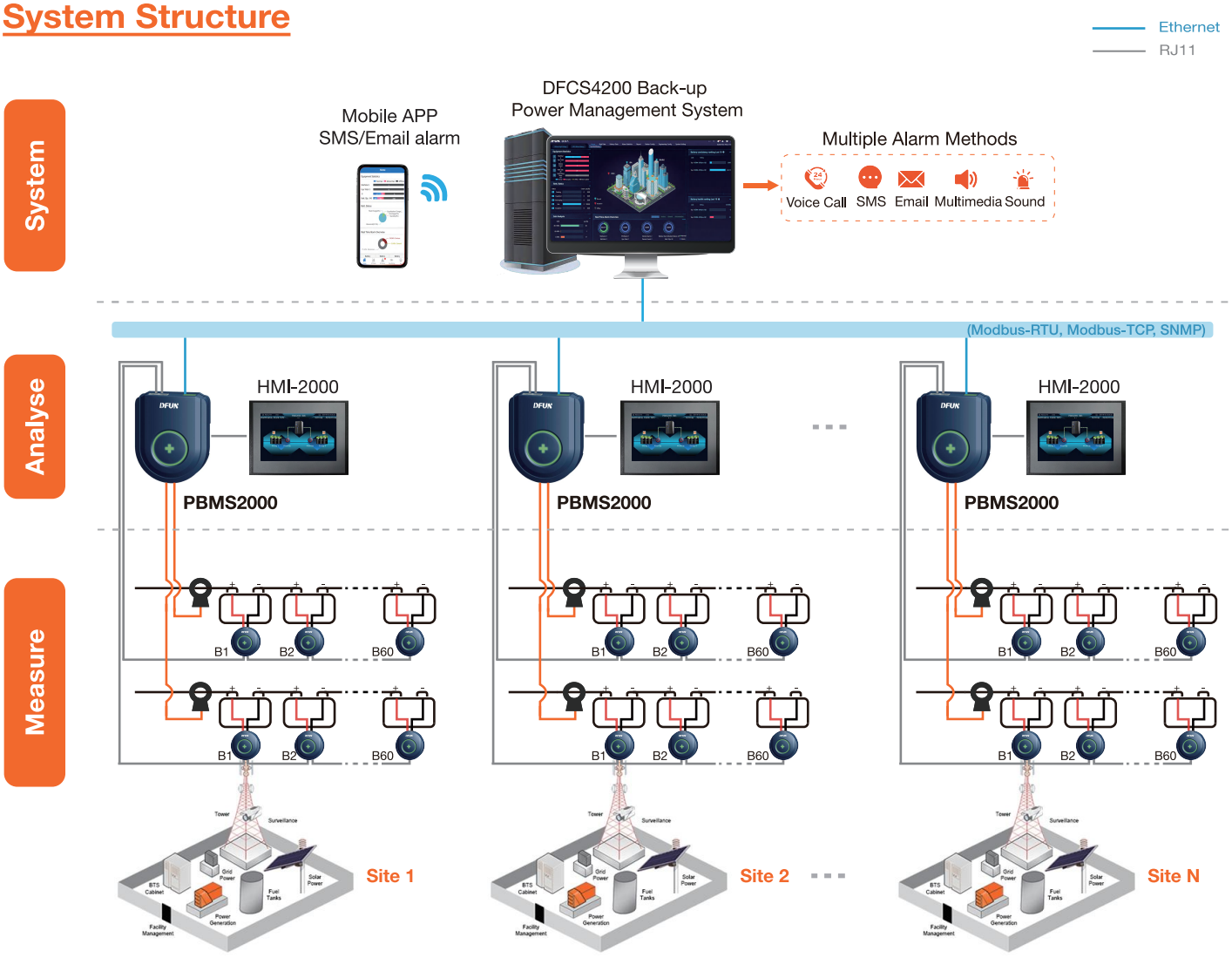
Feature

- Dedicated design for telecommunication 24V or -48V system
- 24/7 hours on-line monitoring & remote alarm
- Cost competitive solution, support Max. 1 string, total 120pcs batteries, or Max. 2 strings, Max. 60pcs batteries each string
- Measure Lead-Acid battery
- Multi communication protocol (Modbus-RTU, MODBUS-TCP, SNMP)
- Comply with IEEE 1188-2005



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System Structure



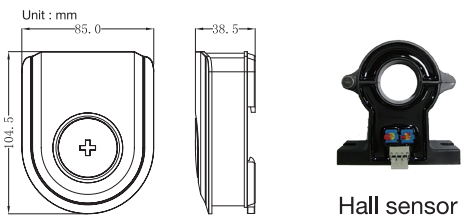
Management Layer

PBMS2000 Battery Management Controller

- Monitor battery string voltage, charge and discharge current
- Calculate battery string SOC
- Auto-balancing
- Auto-sensing for the battery sensor's ID address
- Setpoint alarm (upper limit / lower limit)
- Data collecting
- Data uploading function via build-in RS485 (Modbus-RTU) & Ethernet (Modbus-TCP or SNMP)
- Support 3rd party system
- Local display via HMI
- Accessories:
  1) Hall sensor and cable: range from -1000A ~ +1000A with 2m cable
  2) Communication cable : 5m with RJ11 port



Running status Alarm status



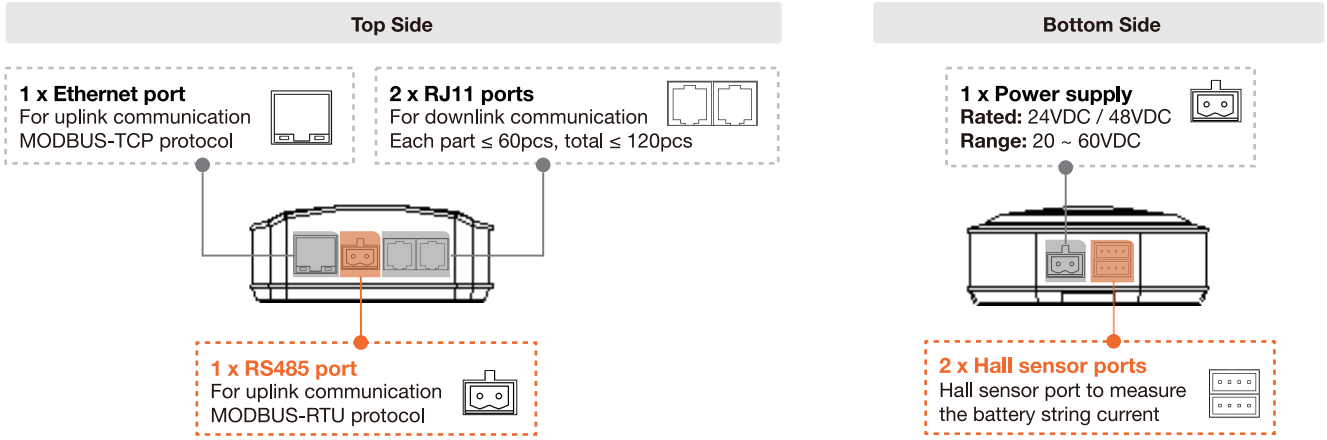
Hall sensor

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Technical Specification

Application		DC power system (Telecom site)
Power Supply		24VDC or 48VDC , range: 20 ~ 60VDC
Measuring	**Voltage**	0~800VDC (accuracy: ±0.5%)
	Current	DC -1000A~+1000A (by hall sensor , accuracy: ±2%)
Power Comsumption		< 2W
Up-link Communication		1 Etherent port (10/100M), Modbus-TCP / SNMP protocol
		1 RS485 port, Modbus-RTU protocol, baudrate: 9600bps, 19200bps, 38400bps (optional)
Down-link Communication		2 RJ11 ports, each port connect Max. 60pcs batteries, total Max. 120pcs batteries

Terminal Introduction



Battery Cell Sensor

PBAT61-02/PBAT61-06/PBAT61-12 Battery Cell Sensor

- PBAT61-02 for 2V battery
- PBAT61-06 for 6V battery
- PBAT61-12 for 12V battery
- Monitor individual battery voltage, internal temperature (negative pole), Impedance(ohmic value)
- Auto-balancing



Running Status

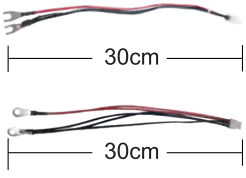


Abnormal Status

Technical Specification

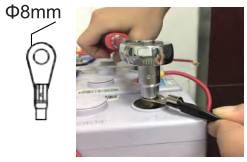
Item	Rated Input Voltage	Measuring Range			
		Voltage	Internal Temperature	Impedance	Power Consumption
PBAT61-02	02V	1.6~2.6V (±0.2%)	-20°C ~ 85°C (±0.5°C)	Range: 0.1~50mΩ Repeatability accuracy: 1.0%±25μΩ Consistency accuracy: 1.5%±25μΩ	Running: <70mW Sleeping: <8mW
PBAT61-06	06V	4.8~10V (±0.2%)			
PBAT61-12	12V	7.5~15.6V (±0.2%)			

Dimension and Installation



	Item	Description
Battery Measuring Cable	Length	30cm
	Terminal & Size	U type, hole diameter: 8mm
		O type, hole diameter: 8mm



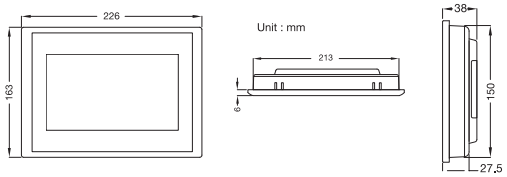


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7-inch Normal HMI (Optional)



- 7-inch touch-screen HMI for display and operation
- **Real-time data inquiry:**
  1) Individual battery voltage, impedance, inner temperature, SOC, SOH
  2) String voltage, charge and discharge current, SOC, balance degree
- **Alarm data inquiry:**
  1) Real-time alarm status and related alarm information (faulty battery ID, date/time, alarm reason, communication status, etc.)
  2) 3000 alarm records for each string battery



Order Information

System Structure		Model	Description	Remark
Management Layer		PBMS2000	Battery Management Controller 48VDC Power Supply	One PBMS2000 can monitor a maximum of 1 string, totaling 120 batteries, or up to 2 strings, each with a maximum of 60 batteries Hall sensor cable: 2m Communication cable: 5m(with RJ11 port)
Battery Cell Sensor		PBAT 61-02	02V battery cell sensor	One per battery cell
		PBAT 61-12	12V battery cell sensor	One per battery cell
Accessory	Cell Sensor Communication Cable	PBAT-COM-40	Cable terminal: RJ11 port, length: 40cm (standard)	
		PBAT-COM-70	Cable terminal: RJ11 port, length: 70cm	
	Battery Measuring Cable	PBAT-C1-U-M8-30	Cable terminal: U type  , hole diameter: 8mm, length: 30cm (standard)	
		PBAT-C1-O-M8-30	Cable terminal: O type  , hole diameter: 8mm, length: 30cm	
	Hall Sensor for PBMS2000	CS050EK1T5	Rated input: 50A	Measure Range: -100A~+100A, Φ20mm
		CS100EK2T5	Rated input: 100A	Measure Range: -200A~+200A, Φ40mm
		CS200EK2T5	Rated input: 200A	Measure Range: -400A~+400A, Φ40mm
		CS300EK2T5	Rated input: 300A	Measure Range: -600A~+600A, Φ40mm
		CS400EK2T5	Rated input: 400A	Measure Range: -800A~+800A, Φ40mm
		CS500EK2T5	Rated input: 500A	Measure Range: -1000A~+1000A, Φ40mm
Optional Module	Local Display & Operation	HMI-2000	7-inch touch-screen HMI for local display & operation	One per PBMS2000 Communication with PBMS2000 via RS485

Notice: If battery terminal is M5, M6, M10, M12, M16, please contact our sales for customize requirements before placing the order

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